

IE416 Final Project

-Techniques and Methods in Human Engineering (2025 Fall)

1. Overview:

The primary objective of this **semester-long course project** is to develop and enhance your skills in analyzing, modeling, and enhancing interactions between humans and other system elements. This will be achieved by applying human engineering techniques and methods, while taking into account the physical and mental demands imposed on human individuals and their overall user experience. **Each project will be undertaken by a group of 2 to 3 students**, fostering collaborative learning and diverse perspectives.

The course project encourages exploration of topics related to "**Human health, safety, performance, and user experience**". Students have full flexibility to select any subject within this theme, and *the topics deal with emerging issues, challenges and opportunities* (human-robot collaboration, human interactions with VR/AR, population aging, digital divide, human-centered AI, human-data interaction, autonomous driving, etc.) *using promising technologies are highly encouraged to try.*

2. Project components, due dates, and grading:

A project proposal, final project oral presentation and final report are due per the schedule and their weightings are shown below.

Component	Weighting	Due Date
Group Forming (2~3 students/group)	0%	Sept 18 (Thur.), 1PM <i>*Group leaders should send the group member information (Name, SID, Email) to the TA via email at: seonghyeok@kaist.ac.kr</i>
Project Proposal (<i>within 5 pages</i>)	25%	Midterm Week
Final Project Presentation (<i>15-min presentation+5-min Q &A</i>)	45%	Weeks 15 & 16

Final Report (Within 10 pages, Conference paper format)	30%	E-submission (TBD)
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3. Final Report:

The report should be in a conference paper format (e-submission is sufficient). The report may be no more than 10 pages including everything (tables, figures, etc.). You will have to be concise! A good project report must include:

- (1) Research background and significance (with extensive literature review)
- (2) Research problem statement and objectives (from observed phenomena or literature review)
- (3) Appropriate research methods and research contents
- (4) Research studies and results
- (5) Conclusions (comments on possible applications, limitations & suggestions for future studies)

4. Remarks:

An extensive, insightful and publishable literature review (see some samples listed below) in the proposed research field can be conducted to replace the course project, if it is difficult to conduct research studies for the proposed topic due to lack of required research resources. ***Such special circumstances require evaluation and approval by the course instructor after reviewing the submitted project proposal.***

- Lee SC, Jeon M, 2022. A systematic review of functions and design features of in-vehicle agents. *International Journal of Human–Computer Studies*, 165, 102864.
- Rappa A, Curtia L, Boldi A, 2021. The human side of human-chatbot interaction: A systematic literature review of ten years of research on text-based chatbots. *International Journal of Human–Computer Studies*, 151, 102630.
- Choi S, Guo L, Kang D and Xiong S, 2017. Exergame technology and interactive interventions for elderly fall prevention: A systematic literature review. *Applied Ergonomics*, 65: 570-581.
- Chen JY and Barnes MJ, 2014. Human–Agent Teaming for Multirobot Control: A Review of Human Factors Issues. *IEEE Transactions on Human-Machine Systems*, 44 (1): 13-29.